

WASP-2b

R-2216

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_120929 · created 2026-08-14T23:14:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456199.6906 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.047
Transit depth (Rp/Rs)^2	2.5 +/- 1.48 [%]
Orbital Inclination (inc)	86.54 +/- 2.86
Airmass coefficient 1 (a1)	3.42 +/- 1.46
Airmass coefficient 2 (a2)	-1.01 +/- 0.44
Scatter in the residuals of the lightcurve fit is	39.75 %
Best Comparison Star	#2 - [159, 337]
Optimal Aperture	2.07
Optimal Annulus	8.52
Transit Duration (day)	0.08 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.047 vs 0.1326 (deviation 0.54σ)
Duration fit vs published	0.08 d vs 0.0737 d (deviation 0.24σ)
Mid-time O–C	-6.5 min
Depth SNR	3.4
Residual scatter	39.75 %

Light curve

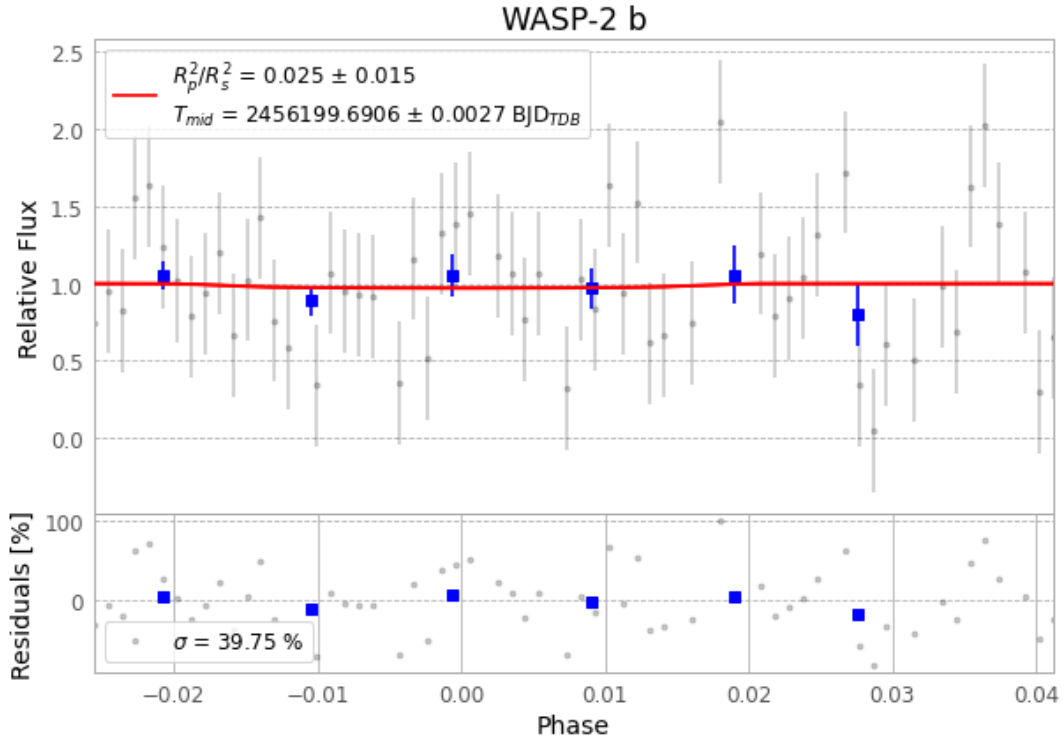


Figure 1 — FinalLightCurve_WASP-2 b_2012-09-28.png (SHA-256 fe59c9b4b93cb382...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [207, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8a0bb40a59608f43...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

71 FITS frames (47 MB), WASP-2120929030913.FITS → WASP-2120929063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120929.log (357ef1a00fb6...), FinalLightCurve_WASP-2 b_2012-09-28.png (fe59c9b4b93c...), FinalLightCurve_WASP-2 b_2012-09-28.csv (4e3981eeceb1...)

Compute resources

Wall time 118.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:14Z.